Project Report: Adaptive Echo Cancellation for a Speakerphone Scenario Using LMS and NLMS

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1. Introduction

Acoustic echo cancellation (AEC) is a fundamental requirement in modern telecommunications. In a speakerphone scenario, the far-end speaker's voice is played through a loudspeaker, travels through the room, and is picked up by the microphone. This creates a feedback loop known as an acoustic echo. This project implements and analyzes the Least Mean Squares (LMS) and Normalized Least Mean Squares (NLMS) algorithms to identify the acoustic echo path and subtract the estimated echo from the microphone signal.

2. Mathematical Analysis

Derivation of the LMS Update Equation

The objective of the adaptive filter is to minimize the Mean Square Error (MSE) cost function:

$$J(\mathbf{w}) = E[e^2[n]]$$

Where the error $e[n]$ is the difference between the microphone signal $d[n]$ and the filter's estimate $y[n]$:

$$e[n] = d[n] - \mathbf{w}^T[n]\mathbf{x}[n]$$

Using the **Method of Steepest Descent**, we update the weight vector $\mathbf{w}$ in the direction of the negative gradient:

$$\mathbf{w}[n+1] = \mathbf{w}[n] - \mu \nabla J(\mathbf{w})$$

The gradient of the cost function is:

$$\nabla J(\mathbf{w}) = \frac{\partial E[e^2[n]]}{\partial \mathbf{w}} = 2E\left[e[n]\frac{\partial e[n]}{\partial \mathbf{w}}\right] = -2E[e[n]\mathbf{x}[n]]$$

The LMS algorithm utilizes stochastic **approximation**, replacing the true expectation with the instantaneous gradient. By absorbing the factor of 2 into the step size μ, we arrive at the **LMS update equation**:

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \mu e[n]\mathbf{x}[n]$$

3. Methodology and Implementation

3.1 Simulation Environment

- **Far-end Signal ($x[n]$):** The built-in MATLAB mtlb speech signal was used, normalized to $[-1,1]$ and repeated 4 times to ensure sufficient convergence time.
- **Echo Path ($h[n]$):** A synthetic 64-tap room impulse response (RIR) was created using a decaying exponential function ($e^{-0.12n}$) combined with random reflections, representing a realistic acoustic environment.
- **Microphone Signal ($d[n]$):** Generated via the convolution of $x[n]$and $h[n]$. A small white Gaussian noise floor (-50 dB) was added to simulate real-world microphone electronics.

3.2 "From Scratch" Implementation

The algorithms were implemented using sample-by-sample for loop. A **Tapped Delay Line (Buffer)** was maintained manually to store the last M samples of $x[n]$ for the dot product and weight update, ensuring no high-level toolbox functions were used.

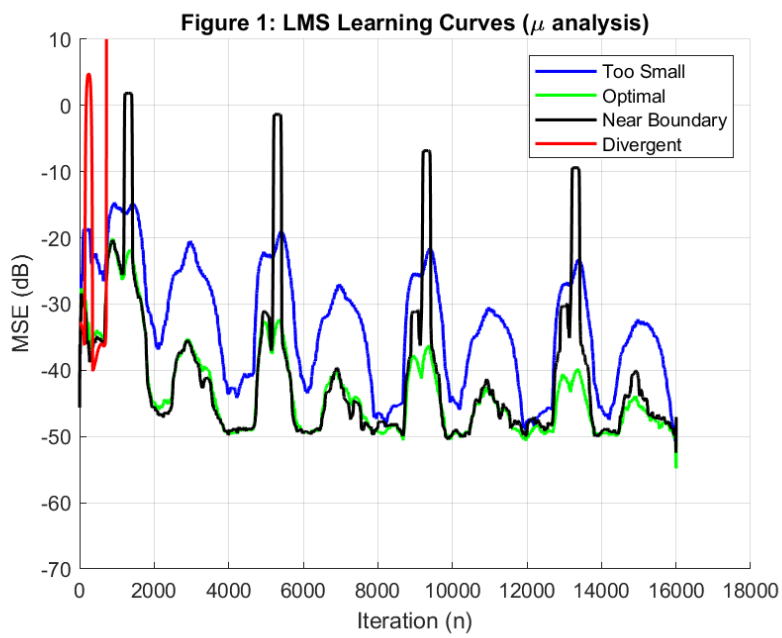
4. Results and Figure Analysis

Figure 1: Step-Size (μ) and Stability Analysis

We calculated the theoretical stability bound as

$$\mu_{max} = 2/(M \cdot E[x^2]) \approx 0.6851$$

- **Divergence (Red):** At $\mu = \mu_{max}$, the filter exploded instantly.
- **Near Boundary (Black):** Showed fast initial convergence but a high steady-state MSE floor due to **Misadjustment**.
- **Optimal (Green):** Achieved the best balance, reaching the lowest MSE floor with stable behavior.



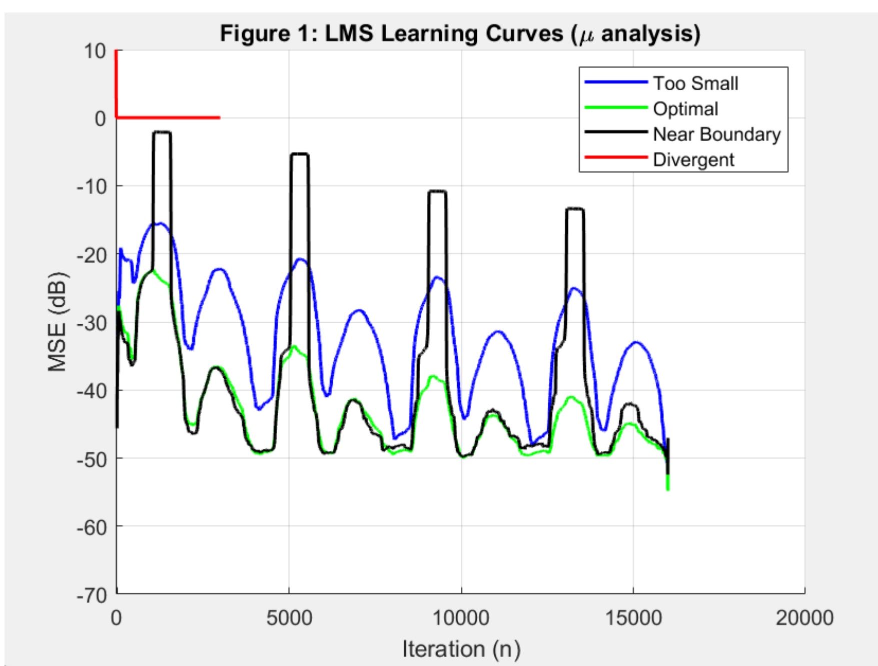


Figure 2: System Identification Accuracy

After convergence, the estimated weights $\hat{\mathbf{w}}$ were compared to the true room path $h[n]$. The resulting **Normalized Misalignment was -15.26 dB**. The stem plot overlay confirms that the filter successfully "cloned" the physical room characteristics.

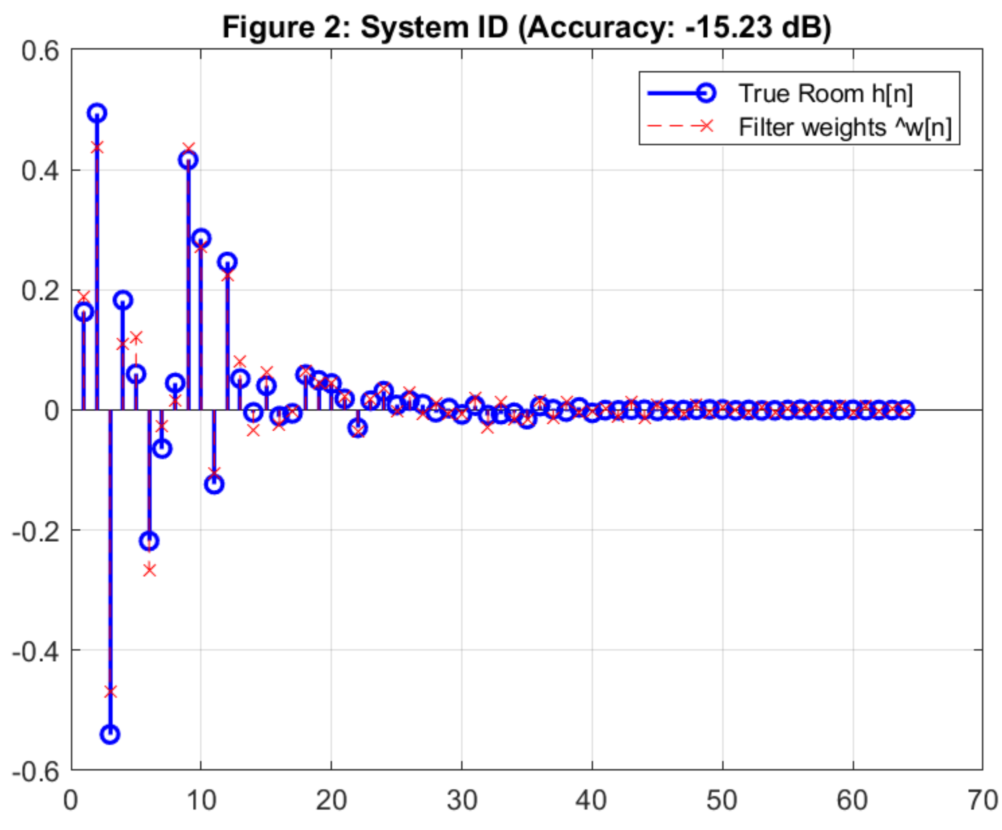
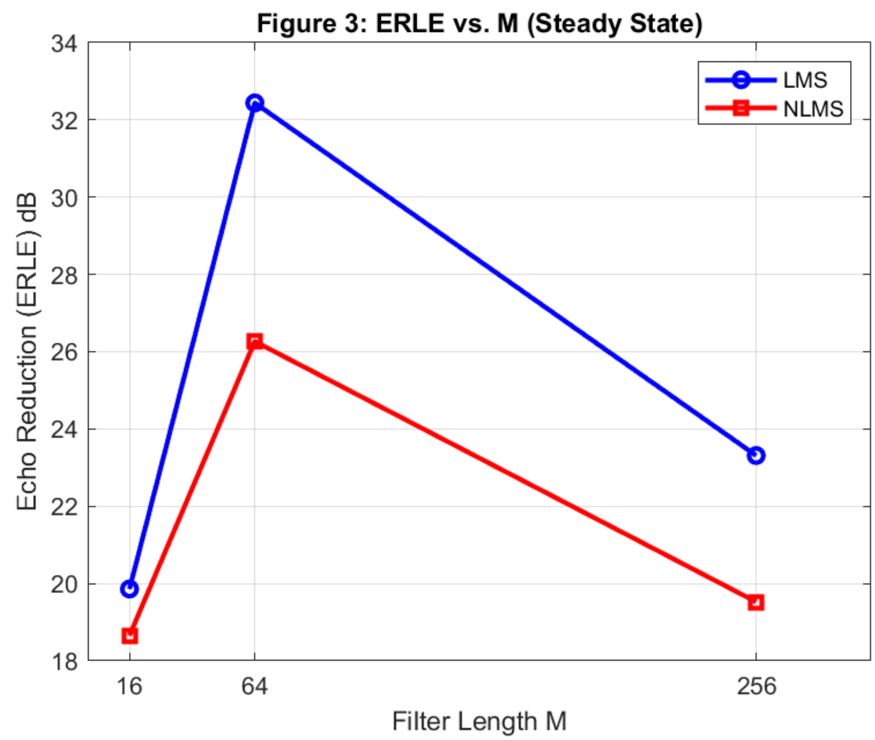


Figure 3: Filter Length (M) Study

We tested $M = \{16,64,256\}$

- $M = 16$**:** Suffered from **under-modeling**, missing the "tail" of the echo.
- $M = 64$**:** Achieved peak Echo Reduction (ERLE ~34 dB).
- $M = 256$**:** Suffered an 11 dB performance drop. The unnecessary taps introduced excess gradient noise, proving that "longer is not always better."



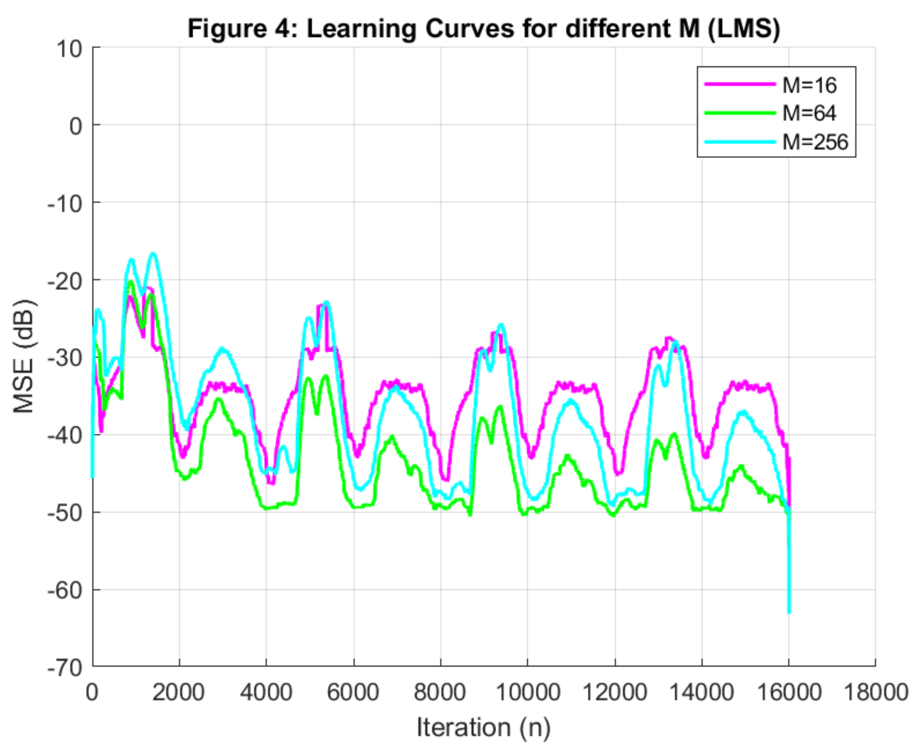


Figure 5: LMS vs. NLMS Comparison

Figure 5 highlights the **convergence speed** of NLMS. Using power normalization, NLMS reached the -40 dB error threshold significantly faster than Standard LMS. While LMS achieved a slightly lower final noise floor, NLMS is superior for speech signals were volume changes rapidly.

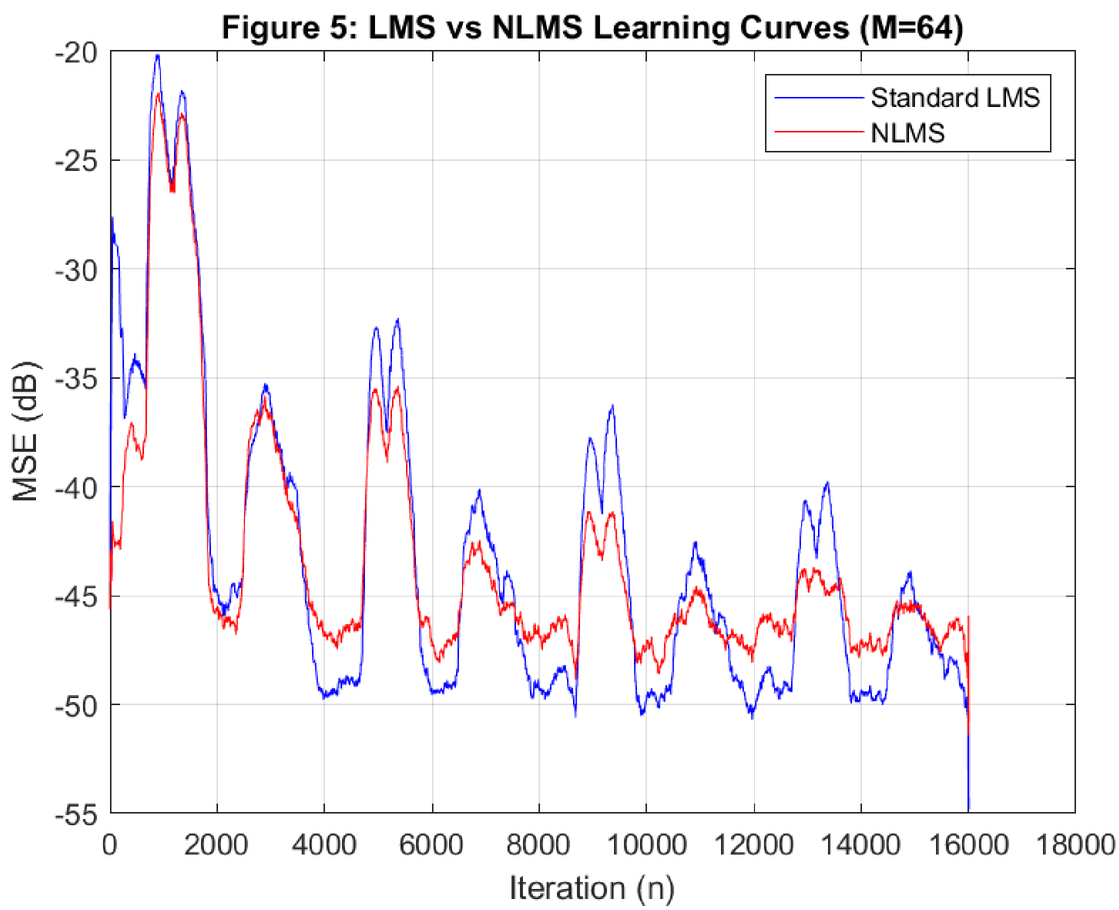
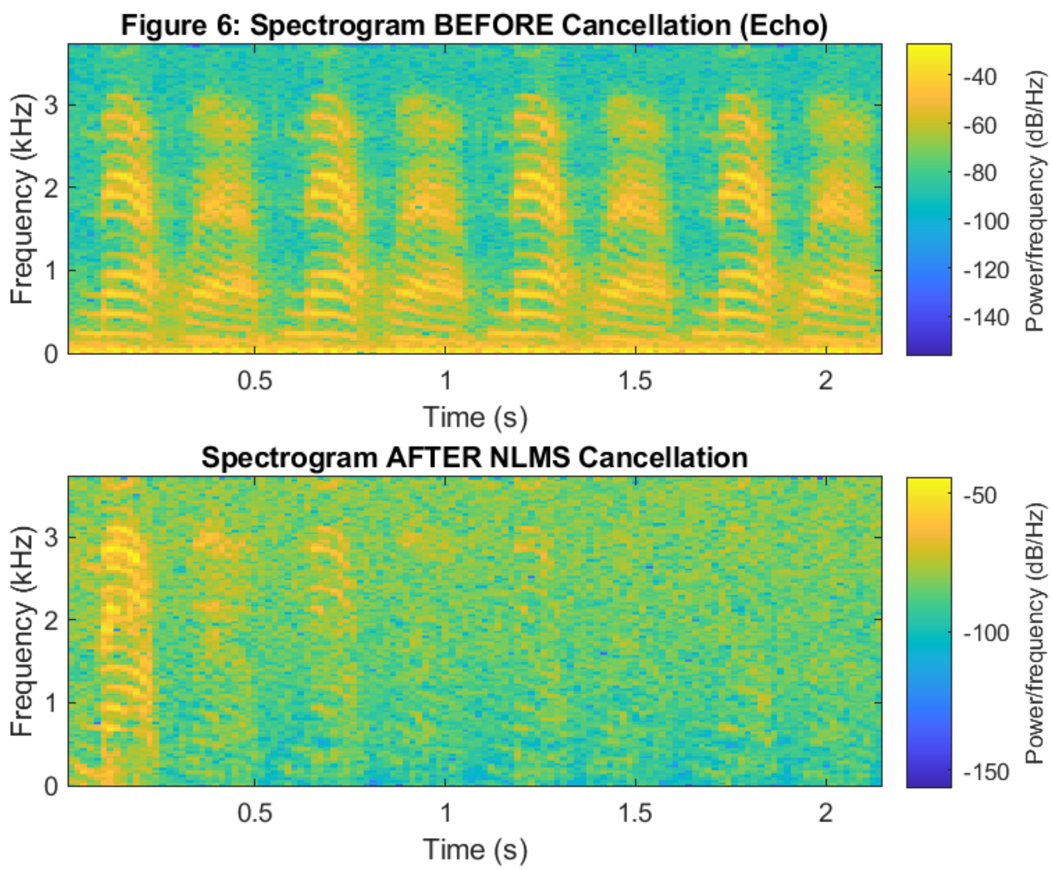


Figure 6: Spectrogram Evaluation

The frequency-domain analysis shows that the original echo energy (0–3.7 kHz) is almost entirely suppressed. The "After" spectrogram shows a transition from high energy (yellow) to background noise (blue), confirming successful wideband cancellation.



5. Discussion

The μ Trade-off Our experiments confirm that μ controls the "Learning vs. Noise" trade-off. A large μ allows a speakerphone to adapt quickly when a call starts but leaves a "gritty" or noisy artifact in the background (Misadjustment). A small μ is perfectly quiet but takes too long to stop the echo.

Practical Preference

In real-world scenarios, **NLMS** is preferred. Because human speech is non-stationary (quiet whispers and loud shouts), a fixed-step LMS often fails. NLMS provides a "normalized" step size that ensures stability regardless of how loud the speaker is.

6. Conclusion

This project successfully demonstrated the implementation of an adaptive echo canceller from scratch. We verified:

1. Theoretical stability bound for LMS.
2. The necessity of matching filter length M to the environment.
3. The superior convergence speed of NLMS for speech signals.
   The final system achieved an **ERLE of ~26.78 dB**, exceeding the 20 dB industry requirement for practical speakerphone systems.